Sound speed and resonance

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use travel time and resonant conditions to interpret sound.

SUCCESS CRITERIA - I can...

1. Account for round-trip distance in an echo.
2. Identify resonance as a response near a natural frequency.
3. Use an appropriate wavelength relationship.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “echo” and “resonance” to explain your model or result. Sentence starters are optional.

echo: a sound received after reflection

resonance: a strong response near a system’s natural frequency

period: time for one complete cycle or orbit

01 Warm-up: bring back an earlier idea

At fixed wave speed, frequency becomes 3 times as large. **Compare** wavelength and typical diffraction through the same opening.

Word help: diffraction = wave spreading around an edge or through an opening

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $d_{echo} = v\,\Delta t/2$; v = 340 m/s.
Echo time is a round trip. Resonance occurs near a natural mode under suitable damping.

Worked reasoning: In an ideal tube open at both ends, the fundamental wavelength is twice its length. For L=0.5 m and sound speed 340 m/s, f=340/1=340 Hz.

Your turn: Sketch the fundamental with displacement antinodes at both ends.

Sentence starter: The ... in my model represents ... . Its label shows ... .

open-open tube

Draw a displacement pattern; label nodes and antinodes.

Marzano check - circle: ① ② ③

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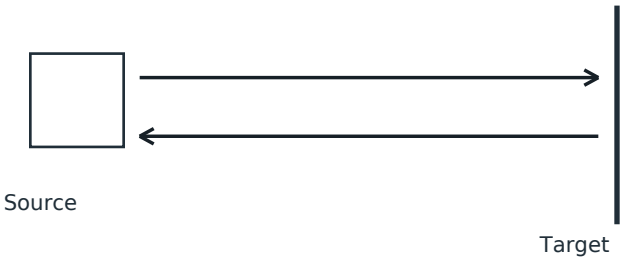
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

An echo returns from a wall.
Draw the outgoing and returning path. Why does sound's total travel distance exceed wall range?

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .



Marzano check - circle: ① ② ③

04 Apply the physics

In air with sound speed 340 m/s, an echo returns after 0.30 s.
Find the wall distance. In a separate instrument model, an open-open pipe is 0.50 m long.
Use λ = 2L and f = v/λ to find its fundamental wavelength and frequency.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Measurement	Path / wavelength	Result
Echo: 0.30 s	Out and back	____ m
Open pipe: 0.50 m	λ = 2L	____ Hz

Marzano check - circle: ① ② ③

05 Explain your reasoning

Why does a resonant response not mean that any object can respond strongly to every frequency?

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A heat engine absorbs Q_{hot}=3500 J and rejects Q_{cold}=2450 J each cycle.
Find work and efficiency.
Label energy entering, leaving and stored.

Word help: efficiency = useful output divided by input

Sentence starter: I found ... [unit]. This value means ... in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A closed-open pipe fundamental has L=0.25 m.
Use λ=4 L and v=340 to find frequency.

Sentence starter: I found … [unit]. This value means … in this situation.

closed-open tube

Draw a displacement pattern; label nodes and antinodes.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 0.85 m tube is closed at one end and open at the other.
Use v=340 m/s and λ=4 L for its fundamental.
Find the first three allowed ideal resonance frequencies.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found … [unit]. This value means … in this situation.

closed-open tube

Draw a displacement pattern; label nodes and antinodes.

Marzano check - circle: 1 2 3